

Seagate

F4Q26 Review: The Demand-Supply Waltz Continues; Raise Estimates & PT Higher

Seagate once again delivered a clear beat-and-raise, one that looks more like a continuation of the durable setup we saw in F3Q26, with results coming in ahead of expectations driven in large part by upside to pricing. On a per-exabyte basis, pricing grew +11% y/y (vs. +4% y/y in F3Q26), with an implied acceleration to at least +20% y/y in F1Q27. Beneath the aggregate trends, management pointed to data center pricing tailwinds from a widening demand-supply gap, while the robust tailwinds in the Edge IoT business reflect opportunistic pricing against a backdrop of very strong price increases for competitors' NAND-based storage products. These increases supported incremental gross margins of 83%+ y/y in F4Q26, along with implied guidance for 88%+ incremental gross margins in F1Q27. Stepping away from pricing, management characterized demand as structurally strengthening, led by sustained cloud buildouts and AI-driven incremental data creation and retention on top of already data-intensive workloads (including video) with no evidence of a near-term slowdown and customer planning horizons extending multiple years, supporting guidance for accelerating revenue growth into FY27. Supply, meanwhile, continues to lag this robust demand, as evident in the vast majority of Nearline exabyte supply already being allocated into CY28 under long-term agreements, with customers actively extending their planning horizons into CY29 and beyond. On new technology platforms, Mozaic 3+ is qualified and in production across all major cloud customers, while Mozaic 4+ is ramping with the two largest CSPs and expanding to additional qualifications. Overall, we are raising our forecasts to bring our pricing assumptions in line with the latest trends and implied guidance, lifting our out-year earnings estimates more than +30% above prior. We reiterate our Overweight rating, driven by unprecedented demand and a favorable pricing environment, and raise our Dec-26 price target to \$1,240 (from \$1,095 prior), driven by a higher earnings outlook.

- F4Q26 (Jun-end) Results: Robust beat driven by revenue and margins tracking ahead of JPMe/consensus.** Revenues tracked to \$3,629 mn (vs. JPMe of \$3,553 mn and consensus of \$3,491 mn), driven by robust data center demand driving data center revenue of \$2,933 mn (+57% y/y). Gross margins tracked to 52.7% (vs. JPMe of 50.9% and consensus of 50.4%), or +570 bps q/q, reflecting the favorable pricing environment driven by a significant gap relative to demand and supply. Opex discipline (8.1% of sales vs JPMe of 8.3%) helped further in driving record operating margins of 44.6% (vs. JPMe of 42.6% and consensus of 41.9%), translating into an EPS beat, with EPS tracking at \$5.71 (vs. JPMe of \$5.25 and consensus of \$5.08).
- F1Q27 (Sep-end) Guide: Better-than-expected guide across revenue, margins and EPS.** Revenue guidance tracked to \$4,000-\$4,200 mn (vs. JPMe of \$3,799 mn and consensus of \$3,792 mn heading into the print), with EPS guidance of \$7.10-\$7.50 (vs. JPMe of \$5.77 and consensus of \$5.81). Operating margin is guided to ~50% (vs. JPMe of 43.6% and consensus of 44.4%), including opex of ~\$300 mn (in-line with JPMe); and an implied gross margin of ~57% (vs. JPMe of 51.5% and consensus of 51.9%) with pricing growth acceleration to at least 20% y/y (vs. JPMe of +12% y/y heading into the print).

Overweight

STX, STX US

Price (28 Jul 26):\$747.30

▲ **Price Target (Dec-27):\$1,240.00**

Prior (Dec-27):\$1,095.00

IT Hardware/ Telecom & Networking Equipment

Joseph Cardoso ^{AC}

(1-212) 622-9036

joseph.cardoso@jpmchase.com

Manmohanpreet Singh

(1-212) 622-4527

manmohanpreet.singh@jpmchase.com

Marc Vitenzon

(1-212) 622-3342

marc.vitenzon@jpmchase.com

Akanksh Chauhan

(1-212) 622-0045

akanksh.chauhan@jpmorgan.com

J.P. Morgan Securities LLC

Key Changes (FYE Jun)

	Prev	Cur	Δ
Adj. EPS - 27E (\$)	28.70	37.50	30.7%
Adj. EPS - 28E (\$)	45.40	60.00	32.1%

Quarterly Forecasts (FYE Jun)

Adj. EPS (\$)	2026A	2027E	2028E
Q1	2.61	7.31	13.10
Q2	3.11	8.67	14.24
Q3	4.10	9.81	15.42
Q4	5.71	11.70	17.22
FY	15.59	37.50	60.00

Style Exposure

Quant	Current	Hist %Rank (1=Top)			
Factors	%Rank	6M	1Y	3Y	5Y
Value	84	76	56	60	23
Growth	2	18	37	91	79
Momentum	3	4	17	97	12
Quality	2	22	7	46	2
Low Vol	62	48	24	28	17
ESGQ	15	14	15	7	-

Sources for: Style Exposure – J.P. Morgan Global Markets Strategy; all other tables are company data and J.P. Morgan estimates.

See page 6 for analyst certification and important disclosures.

Price Performance



	YTD	1m	3m	12m
Abs	171.4%	-17.0%	29.1%	396.7%
Rel	162.9%	-18.0%	25.0%	380.4%

Company Data

Shares O/S (mn)	231
52-week range (\$)	1,145.00-138.30
Market cap (\$ mn)	172,626.30
Exchange rate	1.00
Free float (%)	99.8%
3M ADV (mn)	4.66
3M ADV (\$ mn)	3,981.0
Volatility (90 Day)	80
Index	S&P 500
BBG ANR (Buy Hold Sell)	23 4 0

Key Metrics (FYE Jun)

\$ in millions	FY26A	FY27E	FY28E	FY29E
Financial Estimates				
Revenue	12,195	19,349	27,312	34,862
Adj. EBIT	4,450	10,560	17,020	22,843
Adj. EBITDA	4,708	10,914	17,527	23,496
Adj. net income	3,538	8,695	14,031	18,933
Adj. EPS	15.59	37.50	60.00	80.00
BBG EPS	14.91	28.22	43.28	-
Cashflow from operations	3,674	8,082	13,489	18,639
FCFF	3,105	7,114	12,265	17,244
Margins and Growth				
Revenue Growth Y/Y (%)	34.1%	58.7%	41.2%	27.6%
EBIT margin	36.5%	54.6%	62.3%	65.5%
EBIT Growth Y/Y (%)	109.2%	137.3%	61.2%	34.2%
EBITDA margin	38.6%	56.4%	64.2%	67.4%
EBITDA Growth Y/Y (%)	100.4%	131.8%	60.6%	34.1%
Net margin	29.0%	44.9%	51.4%	54.3%
Adj. EPS growth	92.2%	140.6%	60.0%	33.3%
Ratios				
Adj. tax rate	15.5%	15.9%	16.5%	16.3%
Interest cover	-	-	-	-
Net debt/Equity	0.9	NM	NM	NM
Net debt/EBITDA	0.4	NM	NM	NM
ROE	412.8%	143.6%	85.5%	59.5%
Valuation				
FCFF yield	1.8%	4.1%	7.0%	9.8%
Dividend yield	-	-	-	-
EV/Revenue	14.3	9.0	6.4	5.0
EV/EBITDA	37.1	16.0	10.0	7.4
Adj. P/E	47.9	19.9	12.5	9.3

Summary Investment Thesis and Valuation

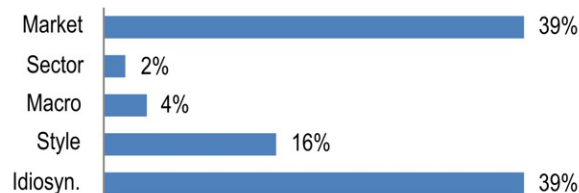
Investment Thesis

We rate Seagate shares Overweight, on account of a very attractive set up for the overall HDD industry, which is positioned to benefit from both strong demand stemming from hyperscaler capex plans as well as pricing tailwinds. STX is ahead in terms of technology transition and is already shipping next-generation HAMR based solutions to data center customers. Near- and medium-term fundamentals continue to improve as the industry is being disciplined in adding new capacity and focusing on profitable growth.

Valuation

We are increasing our December 2027 price target to \$1,240 (vs. \$1,095 prior), based on valuing our 2028E EPS estimate at a ~18x P/E multiple. We believe the target multiple is justified as it is well ahead of the historical multiple driven by robust demand from hyperscaler capex plans, and still conservative relative to some other AI-levered vendors (e.g. Optical) to reflect less aggressive pricing actions.

Performance Drivers



Factors	6M Corr	1Y Corr
Market: MSCI US	0.52	0.62
Sect: Technology	-0.12	-0.18
Ind: Tech Hard Equip	0.25	0.00
Macro:		
Non-Energy Commodity	-0.17	-0.20
US 10yr yield	0.03	-0.15
Crude Oil	-0.12	0.10
Quant Styles:		
Quality	-0.20	-0.33
Momentum	0.33	0.28
Growth	-0.22	-0.12

Investment Thesis, Valuation and Risks

Seagate (Overweight; Price Target: \$1,240.00)

Investment Thesis

We rate Seagate shares Overweight on account of a very attractive set up for the overall HDD industry, which is positioned to benefit from both strong demand stemming from hyperscaler capex plans as well as pricing tailwinds. STX is ahead in terms of technology transition and is already shipping next-generation HAMR based solutions to data center customers. Near- and medium-term fundamentals continue to improve as the industry is being disciplined in adding new capacity and focusing on profitable growth.

Valuation

We are increasing our December 2027 price target to \$1,240 (vs. \$1,095 prior), based on valuing our 2028E EPS estimate at a ~18x P/E multiple. We believe the target multiple is justified as it is well ahead of the historical multiple driven by robust demand from hyperscaler capex plans, and still conservative relative to some other AI-levered vendors (e.g. Optical) to reflect less aggressive pricing actions.

STX P/E-Based Price Target

\$ in mns, except per share amounts

	NTM Qtrs 1-4	2028E
JPM Net Income	8,695	16,625
JPM EPS	\$37.49	\$70.65
P/E Multiple	20x	
JPM P/E Multiple		18x
Implied Equity Value	172,626	291,760
Average Diluted Share Count	231	235
Implied Share Price	\$747.3	\$1,240
Current Value per Share	747.3	747.3
Upside vs. Current		66%
Memo:		
(-) Net Cash/(Debt)	(1,861)	23,116
Enterprise Value	174,487	268,644
JPM EBITDA	10,914	20,685
Implied EV/EBITDA	16.0x	13.0x

Source: Bloomberg Finance L.P. and J.P. Morgan estimates.

Risks to Rating and Price Target

Downside risks to our rating and price target include a slowdown in Cloud capex spending on AI infrastructure, which could present risk to demand growth forecasts for the HDD industry despite the current robust pipeline of investment announcements. Additionally, technology disruptions – including faster-than-expected migration to flash-based storage driven by narrowing cost differentials between HDDs and NAND, and greater progress on

NAND density could present headwinds to our expectations for Seagate shares.

Seagate: Summary of Financials

Income Statement - Annual						Income Statement - Quarterly					
	FY25A	FY26A	FY27E	FY28E	FY29E		1Q27E	2Q27E	3Q27E	4Q27E	
Revenue	9,097	12,195	19,349	27,312	34,862	Revenue	4,119	4,618	5,002	5,610	
COGS	(5,842)	(6,575)	(7,583)	(9,060)	(10,756)	COGS	(1,750)	(1,873)	(1,939)	(2,020)	
Gross profit	3,255	5,620	11,766	18,252	24,106	Gross profit	2,368	2,745	3,063	3,590	
SG&A	(404)	(415)	0	0	0	SG&A	0	0	0	0	
Adj. EBITDA	2,349	4,708	10,914	17,527	23,496	Adj. EBITDA	2,142	2,528	2,854	3,391	
D&A	(222)	(258)	(353)	(507)	(652)	D&A	(74)	(84)	(91)	(103)	
Adj. EBIT	2,127	4,450	10,560	17,020	22,843	Adj. EBIT	2,067	2,443	2,762	3,287	
Net Interest	-	-	-	-	-	Net Interest	-	-	-	-	
Adj. PBT	1,802	4,186	10,342	16,802	22,625	Adj. PBT	2,013	2,389	2,708	3,233	
Tax	(69)	(648)	(1,648)	(2,771)	(3,692)	Tax	(322)	(381)	(431)	(514)	
Minority Interest	-	-	-	-	-	Minority Interest	-	-	-	-	
Adj. Net Income	1,733	3,538	8,695	14,031	18,933	Adj. Net Income	1,691	2,008	2,277	2,719	
Reported EPS	8.11	15.59	42.70	65.27	85.34	Reported EPS	8.62	9.97	11.11	13.00	
Adj. EPS	8.11	15.59	37.50	60.00	80.00	Adj. EPS	7.31	8.67	9.81	11.70	
DPS	-	-	-	-	-	DPS	-	-	-	-	
Payout ratio	-	-	-	-	-	Payout ratio	-	-	-	-	
Shares outstanding	214	227	232	234	237	Shares outstanding	231	232	232	232	
Balance Sheet & CashFlow Statement						Ratio Analysis					
	FY25A	FY26A	FY27E	FY28E	FY29E		FY25A	FY26A	FY27E	FY28E	FY29E
Cash and cash equivalents	891	1,704	7,902	19,075	35,228	Gross margin	35.8%	46.1%	60.8%	66.8%	69.1%
Accounts receivable	959	1,534	2,201	2,996	3,712	EBITDA margin	25.8%	38.6%	56.4%	64.2%	67.4%
Inventories	1,440	1,571	1,917	2,301	2,743	EBIT margin	23.4%	36.5%	54.6%	62.3%	65.5%
Other current assets	2,762	3,517	4,890	6,387	7,846	Net profit margin	19.0%	29.0%	44.9%	51.4%	54.3%
Current assets	3,653	5,221	12,792	25,462	43,074	ROE	(178.3%)	412.8%	143.6%	85.5%	59.5%
PP&E	1,657	2,034	2,628	3,326	4,048	ROA	22.0%	39.3%	61.9%	56.5%	46.5%
LT investments	-	-	-	-	-	ROCE	46.9%	73.2%	92.3%	71.1%	54.0%
Other non current assets	1,492	1,496	1,497	1,497	1,497	SG&A/Sales	4.4%	3.4%	0.0%	0.0%	0.0%
Total assets	8,023	9,972	18,139	31,506	49,840	Net debt/equity	NM	0.9	NM	NM	NM
Short term borrowings	0	185	185	185	185	Net debt/EBITDA	1.7	0.4	NM	NM	NM
Payables	1,604	1,748	2,135	2,563	3,056	P/E (x)	92.2	47.9	19.9	12.5	9.3
Other short term liabilities	1,044	1,194	1,194	1,194	1,194	P/BV (x)	NM	75.4	17.1	7.4	4.2
Current liabilities	2,648	3,127	3,514	3,942	4,435	EV/EBITDA (x)	74.3	37.1	16.0	10.0	7.4
Long-term debt	4,995	3,380	3,380	3,380	3,380	Dividend Yield	-	-	-	-	-
Other long term liabilities	5,828	4,678	4,678	4,678	4,678	Sales/Assets (x)	1.2	1.4	1.4	1.1	0.9
Total liabilities	8,476	7,805	8,192	8,620	9,113	Interest cover (x)	-	-	-	-	-
Shareholders' equity	(453)	2,167	9,947	22,886	40,727	Operating leverage	551.1%	320.7%	234.1%	148.6%	123.8%
Minority interests	-	-	-	-	-	Revenue y/y Growth	38.9%	34.1%	58.7%	41.2%	27.6%
Total liabilities & equity	8,023	9,972	18,139	31,506	49,840	EBITDA y/y Growth	125.7%	100.4%	131.8%	60.6%	34.1%
Net debt/(cash)	4,104	1,861	(4,337)	(15,510)	(31,663)	Tax rate	3.8%	15.5%	15.9%	16.5%	16.3%
Cash flow from operating activities	1,083	3,674	8,082	13,489	18,639	Adj. Net Income y/y Growth	537.2%	104.2%	145.8%	61.4%	34.9%
o/w Depreciation & amortization	251	276	373	527	672	EPS y/y Growth	528.3%	92.2%	140.6%	60.0%	33.3%
o/w Changes in working capital	(965)	(159)	(986)	(1,069)	(966)	DPS y/y Growth	-	-	-	-	-
Cash flow from investing activities	(276)	(525)	(967)	(1,224)	(1,394)						
o/w Capital expenditure	(265)	(569)	(967)	(1,224)	(1,394)						
as % of sales	2.9%	4.7%	5.0%	4.5%	4.0%						
Cash flow from financing activities	(1,274)	(2,337)	(915)	(1,092)	(1,091)						
o/w Dividends paid	(600)	(634)	(675)	(692)	(691)						
o/w Net debt issued/(repaid)	(678)	(1,442)	0	0	0						
o/w Share Repurchases	0	(176)	(240)	(400)	(400)						
Net change in cash	(467)	812	6,199	11,172	16,153						
Adj. Free cash flow to firm	818	3,105	7,114	12,265	17,244						

Source: Company reports and J.P. Morgan estimates.

Note: \$ in millions (except per-share data). Fiscal year ends Jun. o/w - out of which

Analyst Certification: The Research Analyst(s) denoted by an “AC” on the cover of this report certifies (or, where multiple Research Analysts are primarily responsible for this report, the Research Analyst denoted by an “AC” on the cover or within the document individually certifies, with respect to each security or issuer that the Research Analyst covers in this research) that: (1) all of the views expressed in this report accurately reflect the Research Analyst’s personal views about any and all of the subject securities or issuers; and (2) no part of any of the Research Analyst’s compensation was, is, or will be directly or indirectly related to the specific recommendations or views expressed by the Research Analyst(s) in this report. For all Korea-based Research Analysts listed on the front cover, if applicable, they also certify, as per KOFIA requirements, that the Research Analyst’s analysis was made in good faith and that the views reflect the Research Analyst’s own opinion, without undue influence or intervention.

All authors named within this report are Research Analysts who produce independent research unless otherwise specified. In Europe, Sector Specialists (Sales and Trading) may be shown on this report as contacts but are not authors of the report or part of the Research Department.

Important Disclosures

- **Market Maker:** J.P. Morgan Securities LLC makes a market in the securities of Seagate or related entities.
- **Market Maker/ Liquidity Provider:** J.P. Morgan is a market maker and/or liquidity provider in the financial instruments of/related to Seagate or related entities.
- **Beneficial Ownership (1% or more):** J.P. Morgan beneficially owns 1% or more of a class of common equity securities of Seagate or related entities.
- **Client:** J.P. Morgan currently has, or had within the past 12 months, the following entity(ies) as clients: Seagate or related entities.
- **Client/Non-Investment Banking, Securities-Related:** J.P. Morgan currently has, or had within the past 12 months, the following entity(ies) as clients, and the services provided were non-investment-banking, securities-related: Seagate or related entities.
- **Client/Non-Securities-Related:** J.P. Morgan currently has, or had within the past 12 months, the following entity(ies) as clients, and the services provided were non-securities-related: Seagate or related entities.
- **Non-Investment Banking Compensation Received:** J.P. Morgan has received compensation in the past 12 months for products or services other than investment banking from Seagate or related entities.
- **Debt Position:** J.P. Morgan may hold a position in the debt securities of Seagate or related entities, if any.

Company-Specific Disclosures: J.P. Morgan does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. Important disclosures, including price charts and credit opinion history tables (if applicable), are available for compendium reports and all J.P. Morgan–covered companies, and certain non-covered companies, by visiting <https://www.jpm.com/research/disclosures>, calling 1-800-477-0406, or e-mailing research.disclosure.inquiries@jpmorgan.com with your request.

Seagate (STX, STX US) Price Chart



Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Mar 30, 2026. All share prices are as of market close on the previous business day.

The chart(s) show J.P. Morgan's continuing coverage of the stocks; the current analysts may or may not have covered it over the entire period. J.P. Morgan ratings or designations: OW = Overweight, N= Neutral, UW = Underweight, NR = Not Rated

Explanation of Equity Research Ratings, Designations and Analyst(s) Coverage Universe:

J.P. Morgan uses the following rating system: Overweight (over the duration of the price target indicated in this report, we expect this stock will outperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe); Neutral (over the duration of the price target indicated in this report, we expect this stock will perform in line with the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe); and Underweight (over the duration of the price target indicated in this report, we expect this stock will underperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe). NR is Not Rated. In this case, J.P. Morgan has removed the rating and, if applicable, the price target, for this stock because of either a lack of a sufficient fundamental basis or for legal, regulatory or policy reasons. The previous rating and, if applicable, the price target, no longer should be relied upon. An NR designation is not a recommendation or a rating. Some stocks under coverage have a rating but no price target; in these cases, we expect the stock will outperform/perform in line/underperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe of the relevant duration of the region. In our Asia (ex-Australia and ex-India) and U.K. small- and mid-cap Equity Research, each stock's expected total return is compared to the expected total return of a benchmark country market index, not to those Research Analysts' coverage universe. If it does not appear in the Important Disclosures section of this report, the certifying Research Analyst's coverage universe can be found on J.P. Morgan's Research website, <https://www.jpmorganmarkets.com>.

Coverage Universe: Cardoso, Joseph : Amphenol (APH), Arista (ANET), Axon (AXON), CDW (CDW), Calix (CALX), Celestica (CLS), Ceva (CEVA), Ciena (CIEN), Cisco (CSCO), Coherent Corp (COHR), Corning (GLW), Credo (CRDO), Dell Technologies (DELL), Everpure (P), F5 Inc (FFIV), Fabrinet (FN), Flex Ltd (FLEX), Garmin Ltd. (GRMN), HP Inc (HPQ), Hewlett Packard Enterprise (HPE), Ingram Micro (INGM), Insight Enterprises (NSIT), Jabil Inc (JBL), Keysight Technologies (KEYS), Logitech International (LOGI), Lumentum (LITE), Mobileye (MBLY), Motorola Solutions Inc (MSI), NetApp, Inc. (NTAP), Nutanix (NTNX), Qualcomm (QCOM), Seagate (STX), Super Micro (SMCI), TD SYNEX (SNX), Teradyne (TER), Vistance Networks (VISN), Western Digital (WDC), Wolfspeed Inc (WOLF), Xerox Holdings Corp (XRX)

J.P. Morgan Equity Research Ratings Distribution, as of July 04, 2026

	Overweight (buy)	Neutral (hold)	Underweight (sell)
J.P. Morgan Global Equity Research Coverage*	53%	36%	12%
IB clients**	83%	80%	73%
JPMS Equity Research Coverage*	51%	37%	12%
IB clients**	95%	92%	87%

*Please note that the percentages may not add to 100% because of rounding.

**Percentage of subject companies within each of the "buy," "hold" and "sell" categories for which J.P. Morgan has provided investment banking services within the previous 12 months.

For purposes of FINRA ratings distribution rules only, our Overweight rating falls into a buy rating category; our Neutral rating falls into a hold rating category; and our Underweight rating falls into a sell rating category. Please note that stocks with an NR designation are not included in the table above. This information is current as of the end of the most recent calendar quarter.

Equity Valuation and Risks: For valuation methodology and risks associated with covered companies or price targets for covered companies, please see the most recent company-specific research report at <http://www.jpmorganmarkets.com>, contact the primary analyst or your J.P. Morgan representative, or email research.disclosure.inquiries@jpmorgan.com. For material information about the proprietary models used, please see the Summary of Financials in company-specific research reports and the Company Tearsheets, which are available to download on the company pages of our client website, <http://www.jpmorganmarkets.com>. This report also sets out within it the material underlying assumptions used.

History of Investment Recommendations:

A history of J.P. Morgan investment recommendations disseminated during the preceding 12 months can be accessed on the Research & Commentary page of <http://www.jpmorganmarkets.com> where you can also search by analyst name, sector or financial instrument.

Analysts' Compensation: The research analysts responsible for the preparation of this report receive compensation based upon various factors, including the quality and accuracy of research, client feedback, competitive factors, and overall firm revenues.

Other Disclosures

J.P. Morgan is a marketing name for investment banking businesses of JPMorgan Chase & Co. and its subsidiaries and affiliates worldwide.

UK MIFID FICC research unbundling exemption: UK clients should refer to [UK MIFID Research Unbundling exemption](#) for details of J.P. Morgan's implementation of the FICC research exemption and guidance on relevant FICC research categorisation.

All research material made available to clients are simultaneously available on our client website, J.P. Morgan Markets, unless specifically permitted by relevant laws. Not all research content is redistributed, e-mailed or made available to third-party aggregators. For all research material available on a particular stock, please contact your sales representative.

Any long form nomenclature for references to China; Hong Kong; Taiwan; and Macau within this research material are Mainland China; Hong Kong SAR (China); Taiwan (China); and Macau SAR (China).

J.P. Morgan Research may, from time to time, write on issuers or securities targeted by economic or financial sanctions imposed or administered by the governmental authorities of the U.S., EU, UK or other relevant jurisdictions (Sanctioned Securities). Nothing in this report is intended to be read or construed as encouraging, facilitating, promoting or otherwise approving investment or dealing in such Sanctioned Securities. Clients should be aware of their own legal and compliance obligations when making investment decisions.

Any digital or crypto assets discussed in this research report are subject to a rapidly changing regulatory landscape. For relevant regulatory advisories on crypto assets, including bitcoin and ether, please see <https://www.jpmorgan.com/disclosures/cryptoasset-disclosure>.

The author(s) of this research report may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so.

Exchange-Traded Funds (ETFs): J.P. Morgan Securities LLC ("JPMS") acts as authorized participant for substantially all U.S.-listed ETFs. To the extent that any ETFs are mentioned in this report, JPMS may earn commissions and transaction-based compensation in connection with the distribution of those ETF shares and may earn fees for performing other trade-related services, such as securities lending to short sellers of the ETF shares. JPMS may also perform services for the ETFs themselves, including acting as a broker or dealer to the ETFs. In addition, affiliates of JPMS may perform services for the ETFs, including trust, custodial, administration, lending, index calculation and/or maintenance and other services.

Options and Futures related research: If the information contained herein regards options- or futures-related research, such information is available only to persons who have received the proper options or futures risk disclosure documents. Please contact your J.P. Morgan Representative or visit <https://www.theocc.com/components/docs/riskstoc.pdf> for a copy of the Option Clearing Corporation's Characteristics and Risks of Standardized Options or https://www.finra.org/sites/default/files/2020-08/Security_Futures_Risk_Disclosure_Statement_2020.pdf for a copy of the Security Futures Risk Disclosure Statement.

Changes to Interbank Offered Rates (IBORs) and other benchmark rates: Certain interest rate benchmarks are, or may in the future become, subject to ongoing international, national and other regulatory guidance, reform and proposals for reform. For more information, please consult: https://www.jpmorgan.com/global/disclosures/interbank_offered_rates

Private Bank Clients: Where you are receiving research as a client of the private banking businesses offered by JPMorgan Chase & Co. and its subsidiaries ("J.P. Morgan Private Bank"), research is provided to you by J.P. Morgan Private Bank and not by any other division of J.P. Morgan, including, but not limited to, the J.P. Morgan Corporate and Investment Bank and its Global Research division.

Legal entity responsible for the production and distribution of research: The legal entity identified below the name of the Reg AC Research Analyst who authored this material is the legal entity responsible for the production of this research. Where multiple Reg AC Research Analysts authored this material with different legal entities identified below their names, these legal entities are jointly responsible for the production of this research. Where more than one legal entity is listed under an analyst's name, the first legal entity is responsible for the production unless stated otherwise. Research Analysts from various J.P. Morgan affiliates may have contributed to the production of this material but may not be licensed to carry out regulated activities in your jurisdiction (and do not hold themselves out as being able to do so). Unless otherwise stated below in the legal entity disclosures, this material has been distributed by the legal entity responsible for production, or where more than one legal entity is listed under the analyst's name, the first legal entity will be responsible for distribution. If you have any queries, please contact the relevant Research Analyst in your jurisdiction or the entity in your jurisdiction that has distributed this research material.

Legal Entities Disclosures and Country-/Region-Specific Disclosures:

Argentina: JPMorgan Chase Bank N.A Sucursal Buenos Aires is regulated by Banco Central de la República Argentina ("BCRA"- Central Bank of Argentina) and Comisión Nacional de Valores ("CNV"- Argentinian Securities Commission - ALYC y AN Integral N°51).

Australia: J.P. Morgan Securities Australia Limited ("JPMSAL") (ABN 61 003 245 234/AFS Licence No: 238066) is regulated by the Australian Securities and Investments Commission and is a Market Participant of ASX Limited, a Clearing and Settlement Participant of ASX Clear Pty Limited and a Clearing Participant of ASX Clear (Futures) Pty Limited. This material is issued and distributed in Australia by or on behalf of JPMSAL only to "wholesale clients" (as defined in section 761G of the Corporations Act 2001). A list of all financial products covered can be found by visiting <https://www.jpmm.com/research/disclosures>. J.P. Morgan seeks to cover companies of relevance to the domestic and international investor base across all Global Industry Classification Standard (GICS) sectors, as well as across a range of market capitalisation sizes. If applicable, in the course of conducting public side due diligence on the subject company(ies), the Research Analyst team may at times perform such diligence through corporate engagements such as site visits, discussions with company representatives, management presentations, etc. Research issued by JPMSAL has been prepared in accordance with J.P. Morgan Australia's Research Independence Policy which can be found at the following link: [J.P. Morgan Australia - Research Independence Policy](#).

Brazil: Banco J.P. Morgan S.A. is regulated by the Comissão de Valores Mobiliários (CVM) and by the Central Bank of Brazil. Ombudsman J.P. Morgan: 0800-7700847 / 0800-7700810 (For Hearing Impaired) / ouvidoria.jp.morgan@jpmchase.com.

Canada: J.P. Morgan Securities Canada Inc. is a registered investment dealer, regulated by the Canadian Investment Regulatory Organization and the Ontario Securities Commission and is the participating member on Canadian exchanges. This material is distributed in Canada by or on behalf of J.P.Morgan Securities Canada Inc.

Chile: Inversiones J.P. Morgan Limitada is an unregulated entity incorporated in Chile.

China: J.P. Morgan Securities (China) Company Limited has been approved by CSRC to conduct the securities investment consultancy business.

Colombia: Banco J.P. Morgan Colombia S.A. is supervised by the Superintendencia Financiera de Colombia (SFC). Any reference in this material to products or services offered abroad by entities other than the Bank in Colombia is included exclusively for descriptive purposes. Such references do not constitute, and should not be construed as, promotional activity or the provision of financial products or services within Colombian territory, as defined under applicable Colombian regulation.

Dubai International Financial Centre (DIFC): JPMorgan Chase Bank, N.A., Dubai Branch is regulated by the Dubai Financial Services Authority (DFSA) and its registered address is Dubai International Financial Centre - The Gate, West Wing, Level 3 and 9 PO Box 506551, Dubai, UAE. This material has been distributed by JP Morgan Chase Bank, N.A., Dubai Branch to persons regarded as professional clients or market counterparties as defined under the DFSA rules.

European Economic Area (EEA): Unless specified to the contrary, research is distributed in the EEA by J.P. Morgan SE ("JPM SE"), which is authorised as a credit institution by the Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and jointly supervised by the BaFin, the German Central Bank (Deutsche Bundesbank) and the European Central Bank (ECB). JPM SE is a company headquartered in Frankfurt with registered address at TaunusTurm, Taunustor 1, Frankfurt am Main, 60310, Germany. The material has been distributed in the EEA to persons regarded as professional investors (or equivalent) pursuant to Art. 4 para. 1 no. 10 and Annex II of MiFID II and its respective implementation in their home jurisdictions ("EEA professional investors"). This material must not be acted on or relied on by persons who are not EEA professional investors. Any investment or investment activity to which this material relates is only available to EEA relevant persons and will be engaged in only with EEA relevant persons.

Hong Kong: J.P. Morgan Securities (Asia Pacific) Limited (CE number AAJ321) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission in Hong Kong, and J.P. Morgan Broking (Hong Kong) Limited (CE number AAB027) is regulated by the Securities and Futures Commission in Hong Kong. JP Morgan Chase Bank, N.A., Hong Kong Branch (CE Number AAL996) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission, is organized under the laws of the United States with limited liability. Where the distribution of this material is a regulated activity in Hong Kong, the material is distributed in Hong Kong by or through J.P. Morgan Securities (Asia Pacific) Limited and/or J.P. Morgan Broking (Hong Kong) Limited.

India: J.P. Morgan India Private Limited (Corporate Identity Number - U67120MH1992FTC068724), having its registered office at J.P. Morgan Tower, Off. C.S.T. Road, Kalina, Santacruz - East, Mumbai - 400098, is registered with the Securities and Exchange Board of India (SEBI) as a 'Research Analyst' having registration number INH000001873. J.P. Morgan India Private Limited is also registered with SEBI as a member of the National Stock Exchange of India Limited and the Bombay Stock Exchange Limited (SEBI Registration Number - INZ000239730) and as a Merchant Banker (SEBI Registration Number - MB/INM000002970). Telephone: 91-22-6157 3000, Facsimile: 91-22-6157 3990 and Website: <http://www.jpmpi.com>. JPMorgan Chase Bank, N.A. - Mumbai Branch is licensed by the Reserve Bank of India (RBI) (Licence No. 53/ Licence No. BY.4/94; SEBI - IN/CUS/014/ CDSL : IN-DP-CDSL-444-2008/ IN-DP-NSDL-285-2008/ INBI00000984/ INE231311239) as a Scheduled Commercial Bank in India, which is its primary license allowing it to carry on Banking business in India and other activities, which a Bank branch in India are permitted to undertake. For non-local research material, this material is not distributed in India by J.P. Morgan India Private Limited. Compliance Officer: Ashutosh Sharma; ashutosh.j.sharma@jpmchase.com; +912261575002. Grievance Officer: Ramprasad K, jpmpi.research.feedback@jpmorgan.com; +912261573000. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Please visit [Terms and Conditions and Most Important Terms and Conditions \(MITC\)](#). The annual Compliance audit report is available at <http://www.jpmpi.com/#research>.

Indonesia: PT J.P. Morgan Sekuritas Indonesia is a member of the Indonesia Stock Exchange and is registered and supervised by the Otoritas Jasa Keuangan (OJK).

Korea: J.P. Morgan Securities (Far East) Limited, Seoul Branch, is a member of the Korea Exchange (KRX). JPMorgan Chase Bank, N.A., Seoul Branch, is licensed as a branch office of foreign bank (JPMorgan Chase Bank, N.A.) in Korea. Both entities are regulated by the Financial Services Commission (FSC) and the Financial Supervisory Service (FSS). For non-macro research material, the material is distributed in Korea by or through J.P. Morgan Securities (Far East) Limited, Seoul Branch.

Japan: JPMorgan Securities Japan Co., Ltd. and JPMorgan Chase Bank, N.A., Tokyo Branch are regulated by the Financial Services Agency in Japan.

Malaysia: This material is issued and distributed in Malaysia by JPMorgan Securities (Malaysia) Sdn Bhd (18146-X), which is a Participating Organization of Bursa Malaysia Berhad and holds a Capital Markets Services License issued by the Securities Commission in Malaysia.

Mexico: J.P. Morgan Casa de Bolsa, S.A. de C.V., J.P. Morgan Grupo Financiero is member of the Mexican Stock Exchange ("Bolsa Mexicana de Valores") and the Institutional Stock Exchange ("Bolsa Institucional de Valores"), and it is authorized to act as a broker dealer by the National Banking and Securities Exchange Commission ("Comisión Nacional Bancaria y de Valores").

New Zealand: This material is issued and distributed by JPMSAL in New Zealand only to "wholesale clients" (as defined in the Financial Markets Conduct Act 2013). JPMSAL is registered as a Financial Service Provider under the Financial Service providers (Registration and Dispute Resolution) Act of 2008.

Philippines: J.P. Morgan Securities Philippines Inc. is a Trading Participant of the Philippine Stock Exchange and a member of the Securities Clearing Corporation of the Philippines and the Securities Investor Protection Fund. It is regulated by the Securities and Exchange Commission.

Singapore: This material is issued and distributed in Singapore by or through J.P. Morgan Securities Singapore Private Limited (JPMSS) [MDDI (P) 057/08/2025 and Co. Reg. No.: 199405335R], which is a member of the Singapore Exchange Securities Trading Limited, and/or JPMorgan Chase Bank, N.A., Singapore branch (JPMCB Singapore), both of which are regulated by the Monetary Authority of Singapore. This material is issued and distributed in Singapore only to accredited investors, expert investors and institutional investors, as defined in Section 4A of the Securities and Futures Act, Cap. 289 (SFA). This material is not intended to be issued or distributed to any retail investors or any other investors that do not fall into the classes of "accredited investors," "expert investors" or "institutional investors," as defined under Section 4A of the SFA. Recipients of this material in Singapore are to contact JPMSS or JPMCB Singapore in respect of any matters arising from, or in connection with, the material.

South Africa: J.P. Morgan Equities South Africa Proprietary Limited and JPMorgan Chase Bank, N.A., Johannesburg Branch are members of the Johannesburg Securities Exchange and are regulated by the Financial Services Conduct Authority (FSCA).

Taiwan: J.P. Morgan Securities (Taiwan) Limited is a participant of the Taiwan Stock Exchange (company-type) and regulated by the Taiwan Securities and Futures Bureau. Material relating to equity securities is issued and distributed in Taiwan by J.P. Morgan Securities (Taiwan) Limited, subject to the license scope and the applicable laws and the regulations in Taiwan. **To the extent that J.P. Morgan Securities (Taiwan) Limited produces research materials on securities not listed on the Taiwan Stock Exchange or Taipei Exchange ("Non-Taiwan Listed Securities"), these materials shall not constitute securities recommendations for the purpose of applicable Taiwan regulations, and, for the avoidance of doubt, J.P. Morgan Securities (Taiwan) Limited does not act as broker for Non-Taiwan Listed Securities.** According to Paragraph 2, Article 7-1 of Operational Regulations Governing Securities Firms Recommending Trades in Securities to Customers (as amended or supplemented) and/or other applicable laws or regulations, please note that the recipient of this material is not permitted to engage in any activities in connection with the material that may give rise to conflicts of interests, unless otherwise disclosed in the "Important Disclosures" in this material.

Thailand: This material is issued and distributed in Thailand by JPMorgan Securities (Thailand) Ltd., which is a member of the Stock Exchange of Thailand and is regulated by the Ministry of Finance and the Securities and Exchange Commission. The registered address is 548 One City Center Building, 50th Floor, Ploenchit Road, Lumpini, Pathum Wan, Bangkok 10330.

UK: Research is produced in the UK by J.P. Morgan Securities plc ("JPMS plc") which is a member of the London Stock Exchange and is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority or J.P. Morgan Markets Limited ("JPML Ltd") which is authorised and regulated by the Financial Conduct Authority. Unless specified to the contrary, this material is distributed in the UK by JPMS plc and is directed in the UK only to: (a) persons having professional experience in matters relating to investments falling within article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) (Order) 2005 ("the FPO"); (b) persons outlined in article 49 of the FPO (high net worth companies, unincorporated associations or partnerships, the trustees of high value trusts, etc.); or (c) any persons to whom this communication may otherwise lawfully be made; all such persons being referred to as "UK relevant persons". This material must not be acted on or relied on by persons who are not UK relevant persons. Any investment or investment activity to which this material relates is only available to UK relevant persons and will be engaged in only with UK relevant persons. A description of J.P. Morgan EMEA's policy for prevention and avoidance of conflicts of interest related to the production of Research can be found at the following link: [J.P. Morgan EMEA - Research Independence Policy](#).

U.S.: J.P. Morgan Securities LLC ("JPMS") is a member of the NYSE, FINRA, SIPC, and the NFA. JPMorgan Chase Bank, N.A. is a member of the FDIC. Material published by non-U.S. affiliates is distributed in the U.S. by JPMS who accepts responsibility for its content.

General: Additional information is available upon request. The information in this material has been obtained from sources believed to be reliable. While all reasonable care has been taken to ensure that the facts stated in this material are accurate and that the forecasts, opinions and expectations contained herein are fair and reasonable, JPMorgan Chase & Co. or its affiliates and/or subsidiaries (collectively J.P. Morgan) make no representations or warranties whatsoever to the completeness or accuracy of the material provided, except with respect to any disclosures relative to J.P. Morgan and the Research Analyst's involvement with the issuer that is the subject of the material. Accordingly, no reliance should be placed on the accuracy, fairness or completeness of the information contained in this material. There may be certain discrepancies with data and/or limited content in this material as a result of calculations, adjustments, translations to different languages, and/or local regulatory restrictions, as applicable. These discrepancies should not impact the overall investment analysis, views and/or recommendations of the subject company(ies) that may be discussed in the material. Artificial intelligence tools may have been used in the preparation of this material, including assisting in data analysis, pattern recognition, and content drafting for research material. J.P. Morgan accepts no liability whatsoever for any loss arising from any use of this material or its contents, and neither J.P. Morgan nor any of its respective directors, officers or employees, shall be in any way responsible for the contents hereof, apart from the liabilities and responsibilities that may be imposed on them by the relevant regulatory authority in the jurisdiction in question, or the regulatory regime thereunder. Opinions, forecasts or projections contained in this material represent J.P. Morgan's current opinions or judgment as of the date of the material only and are therefore subject to change without notice. Periodic updates may be provided on companies/industries based on company-specific developments or announcements, market conditions or any other publicly available information. There can be no assurance that future results or events will be consistent with any such opinions, forecasts or projections, which represent only one possible outcome. Furthermore, such opinions, forecasts or projections are subject to certain risks, uncertainties and assumptions that have not been verified, and future actual results or events could differ materially. The value of, or income from, any investments referred to in this material may fluctuate and/or be affected by changes in exchange rates. All pricing is indicative as of the close of market for the securities discussed, unless otherwise stated. Past performance is not indicative of future results.

Accordingly, investors may receive back less than originally invested. This material is not intended as an offer or solicitation for the purchase or sale of any financial instrument. The opinions and recommendations herein do not take into account individual client circumstances, objectives, or needs and are not intended as recommendations of particular securities, financial instruments or strategies to particular clients. This material may include views on structured securities, options, futures and other derivatives. These are complex instruments, may involve a high degree of risk and may be appropriate investments only for sophisticated investors who are capable of understanding and assuming the risks involved. The recipients of this material must make their own independent decisions regarding any securities or financial instruments mentioned herein and should seek advice from such independent financial, legal, tax or other adviser as they deem necessary. J.P. Morgan may trade as a principal on the basis of the Research Analysts' views and research, and it may also engage in transactions for its own account or for its clients' accounts in a manner inconsistent with the views taken in this material, and J.P. Morgan is under no obligation to ensure that such other communication is brought to the attention of any recipient of this material. Others within J.P. Morgan, including Strategists, Sales staff and other Research Analysts, may take views that are inconsistent with those taken in this material. Employees of J.P. Morgan not involved in the preparation of this material may have investments in the securities (or derivatives of such securities) mentioned in this material and may trade them in ways different from those discussed in this material. This material is not an advertisement for or marketing of any issuer, its products or services, or its securities in any jurisdiction.

Confidentiality and Security Notice: This transmission may contain information that is privileged, confidential, legally privileged, and/or exempt from disclosure under applicable law. If you are not the intended recipient, you are hereby notified that any disclosure, copying, distribution, or use of the information contained herein (including any reliance thereon) is STRICTLY PROHIBITED. Although this transmission and any attachments are believed to be free of any virus or other defect that might affect any computer system into which it is received and opened, it is the responsibility of the recipient to ensure that it is virus free and no responsibility is accepted by JPMorgan Chase & Co., its subsidiaries and affiliates, as applicable, for any loss or damage arising in any way from its use. If you received this transmission in error, please immediately contact the sender and destroy the material in its entirety, whether in electronic or hard copy format. This message is subject to electronic monitoring: <https://www.jpmorgan.com/disclosures/email>

MSCI: Certain information herein ("Information") is reproduced by permission of MSCI Inc., its affiliates and information providers ("MSCI") ©2026. No reproduction or dissemination of the Information is permitted without an appropriate license. MSCI MAKES NO EXPRESS OR IMPLIED WARRANTIES (INCLUDING MERCHANTABILITY OR FITNESS) AS TO THE INFORMATION AND DISCLAIMS ALL LIABILITY TO THE EXTENT PERMITTED BY LAW. No Information constitutes investment advice, except for any applicable Information from MSCI ESG Research. Subject also to [msci.com/disclaimer](https://www.msci.com/disclaimer)

Sustainalytics: Certain information, data, analyses and opinions contained herein are reproduced by permission of Sustainalytics and: (1) includes the proprietary information of Sustainalytics; (2) may not be copied or redistributed except as specifically authorized; (3) do not constitute investment advice nor an endorsement of any product or project; (4) are provided solely for informational purposes; and (5) are not warranted to be complete, accurate or timely. Sustainalytics is not responsible for any trading decisions, damages or other losses related to it or its use. The use of the data is subject to conditions available at <https://www.sustainalytics.com/legal-disclaimers>. ©2026 Sustainalytics. All Rights Reserved.

"Other Disclosures" last revised July 04, 2026.

Copyright 2026 JPMorgan Chase & Co. All rights reserved. This material or any portion hereof may not be reprinted, sold or redistributed without the written consent of J.P. Morgan. It is strictly prohibited to use or share without prior written consent from J.P. Morgan any research material received from J.P. Morgan or an authorized third-party ("J.P. Morgan Data") in any third-party artificial intelligence ("AI") systems or models when such J.P. Morgan Data is accessible by a third-party.

Completed 29 Jul 2026 01:39 AM EDT

Disseminated 29 Jul 2026 01:39 AM EDT